

9.3 Writing a Quadratic Function From a Table

Lesson Introduction

 Benchmarks	MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.		 Learning Targets	- I can write a quadratic function to represent the relationship between two quantities from a table. - I can rewrite a quadratic equation from vertex form to standard form. - I can solve for a variable of interest.	
 Benchmark Coherence	Building On MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.			Working Toward MA.912.AR.3.4 Write a quadratic function to represent the relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context, clarification 3.	
 Essential Questions	- How can you tell if a table represents a quadratic relationship? - What key information about a quadratic function can be determined from a table? - How do you write a quadratic function from a table?				
 Lesson Narrative	In this lesson, students will continue working on writing a quadratic function. Students will discover how to write the equation of a quadratic function when given a table. The activities in this lesson provide instruction on getting key information from a table and using that information to write a quadratic function in both vertex and standard form.				
 Component Summary	9.3.1 Warm-Up (~5 min.)	Graph and table of value matching (<i>SE p. 71</i>)	9.3.3 Guided Practice (15 min.)	Practice writing a quadratic function using a vertex and point from a table (<i>SE p. 73</i>)	
	9.3.2 Exploration (10 min.)	Explore and discover how to identify the vertex from a table (<i>SE p. 72</i>)	9.3.4 Your Turn! (10 min.)	Independently practice writing a quadratic equation in vertex and/or standard form given a table of values (<i>SE p. 75</i>)	
	9.3.5 Wrap-Up (~5 min.)	Ticket Out the Door on writing the equation of a quadratic function in vertex and standard form (<i>SE p. 76</i>)			
 Resources	Glossary	Materials	Additional Preparation		
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> - Quadratic equation x-intercept	N/A	N/A		

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may assume that any “middle” value in a table or values represents the vertex. See 9.3.5 Wrap-Up for an example (students may identify (4,2) as the vertex).	Provide an opportunity for students to explore and make connections as they develop a working understanding of using a table of values to find the vertex. Monitor for specific misconceptions and use those observations to inform possible small groups for 9.3.3 Guided Practice.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Think-Pair-Share	MA.K12.MTR.4.1: Discussion and Reflection
	9.3.2 Exploration provides students with a collaborative opportunity to work with other students in order to explore how to identify the vertex from a table. Students will be able to think about the questions on their own, pair up with a partner, and share their thinking to learn from one another and come to a shared understanding.	9.3.3 Guided Practice provides questions that ask students to discuss the methods and strategies they chose in order to compare them with one another.
Utilize the Collect and Display routine during 9.3.3 Guided Practice with a large-scale chart paper or other method for posting work for the class to see. Use this method to provide a visual aid for students who are visual learners. - Students can work in groups and create their own anchor chart. - This can be teacher-led with a small group based on common misconceptions. - Alternatively, you could use the chart paper as an opportunity for synthesis, acting as a “Bring It Together” activity between Guided Practice and Your Turn!		
Lesson Notes:		

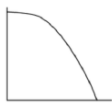
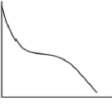
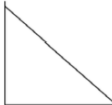
9.3.1 Warm-Up: Words, Tables, and Graphs – Part 1

Component Overview: Students will determine which graph is the best representation of the given table of values. This activity is designed to be done independently. The working knowledge of graphs may impact the depth of description that students are able to do.



9.3.1 Warm-Up: Words, Tables, and Graphs - Part 1

- Match each of the following three graphs with the appropriate table of values.

Graphs	Table of Values														
I. 	A. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>400</td><td>320</td><td>240</td><td>160</td><td>80</td><td>0</td></tr></table>	x	0	1	2	3	4	5	y	400	320	240	160	80	0
x	0	1	2	3	4	5									
y	400	320	240	160	80	0									
II. 	B. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>400</td><td>384</td><td>336</td><td>256</td><td>144</td><td>0</td></tr></table>	x	0	1	2	3	4	5	y	400	384	336	256	144	0
x	0	1	2	3	4	5									
y	400	384	336	256	144	0									
III. 	C. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>400</td><td>253</td><td>218</td><td>216</td><td>181</td><td>34</td></tr></table>	x	0	1	2	3	4	5	y	400	253	218	216	181	34
x	0	1	2	3	4	5									
y	400	253	218	216	181	34									

I & B; II & C; III & A

- How did you use each table to determine the graph it matched to?

Answers will vary. The y -values showed how quickly the graph was decreasing. I was able to see which one was linear, and then I used the rate at which the other two were decreasing to match to the right graph.



Challenge students to describe a real-world scenario that could represent the graph and table of values after matching. Students will be asked to do this in the next lesson. Students who are already ahead can set the groundwork for what will be done in the Warm-Up of the next lesson.



Content here will appear again with a different task in the next lesson. Have students retain their work, approaches, and solutions to compare and connect for the next lesson.

9.3.2 Exploration: Identifying the Vertex From a Table

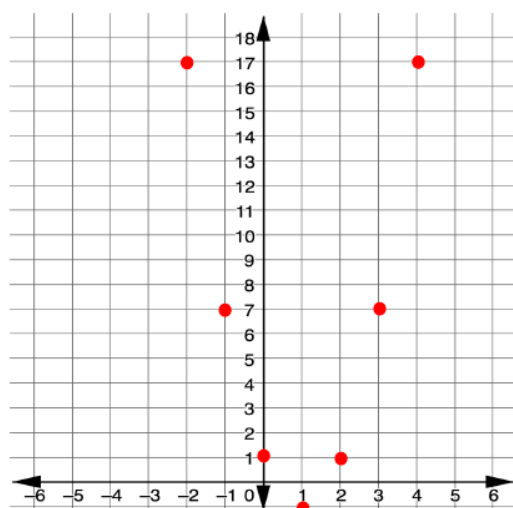
Component Overview: Students will work with a partner to explore and discover how to identify the vertex from a table. This is an exploration activity, so students should be paired together or assigned to collaborative working groups. Allow students to complete questions 1-10 on their own with their partner/group. After a set time is over, review each question with the students as a class.



9.3.2 Exploration: Identifying the Vertex from a Table

- Plot the points from the table on the graph.

x	-2	-1	0	1	2	3	4
y	17	7	1	-1	1	7	17



Vertex = (1, -1)

- Label the vertex.
Answer on graph above.
- Discuss with your partner what change happens at the vertex in the graph. Summarize your discussion.
Answers will vary. At the vertex, the graph changes direction. The y -values are decreasing before the vertex, but after the vertex, the y -values begin to increase.
- Thinking of the change that you observed in the graph, review the table and look for a similar change.
At the vertex, (1, -1), the y -values stop decreasing and begin increasing.



There may be students who have difficulty plotting points correctly. Consider your student pairings so those students can receive additional support. If any pairings have prior knowledge deficits, consider working with these groups first in order to help them get off to a strong start for this activity.



Does the parabola open up or down? What does this tell you about the coefficient of the x^2 term?



Listen for students using correct language and key features. If students are struggling with language, then refer them to the key features learned in Unit 8.

9.3.2 Exploration: Identifying the Vertex From a Table (continued)

5. Discuss with your partner how the vertex can be identified from the table without plotting the points. Summarize your discussion using the key feature of increasing/decreasing.

Answers will vary. The vertex can be identified from the table by looking at the y -values and finding where the values change from decreasing to increasing.

6. On the graph, label two points that are equidistant from the vertex.

Answers will vary. (0,1) & (2,1). (-1,7) & (3,7).

7. In the table, draw a square around the two points labeled on the graph.

Answers will vary.

x	-2	-1	0	1	2	3	4
y	17	7	1	-1	1	7	17

x	-2	-1	0	1	2	3	4
y	17	7	1	-1	1	7	17

8. Complete the statement.

Two points that are equidistant from the vertex can be identified by matching points that have the same ...

y-values

9. In the table, circle two additional points that are equidistant from the vertex.

Answers will vary.

x	-2	-1	0	1	2	3	4
y	17	7	1	-1	1	7	17

x	-2	-1	0	1	2	3	4
y	17	7	1	-1	1	7	17

10. Discuss with your partner if there is enough information from the table to write the equation of the quadratic function in vertex form. Justify your choice by explaining your reasoning.

Yes, there is enough information. Justifications will vary. There is enough information because the vertex is given and an additional point is given. There is enough information because there is a vertex and additional points.



What does it mean for points to be equidistant? What key feature other than the vertex helps to determine which points are equidistant?



Question 8 scaffolds student understanding of equidistance from the vertex on the parabola. Connect the multiple representations for students as well as the connection to the axis of symmetry.



What key feature is absolutely necessary in order to write a quadratic function in vertex form?

9.3.3 Guided Practice: Writing a Quadratic Function Using a Vertex and Point From a Table

Component Overview: Students will be provided with a Guided Practice opportunity for writing a quadratic function using a vertex and point from a table. For question 1 part a, introduce students to the values and remind them of the vertex and release to have them try and find it. For part b, show students how to do one point, modeling all the steps, and then release students to do part b on their own and part c with a partner. For question 2, allow students to try on their own and then come back together to review. Repeat this for question 3.



9.3.3 Guided Practice: Writing a Quadratic Function Using a Vertex and Point from a Table

1. The table of values for a quadratic function is provided.

x	-7	-5	-3	-1	1	3
$f(x)$	-7	5	9	5	-7	-27

- a. Determine the vertex of the function.
 $(-3, 9)$
- b. Using the vertex, (h, k) , and any point, $(x, f(x))$, in the table, fill in the steps for writing the equation for the quadratic function.

Step	Equation
Substitute values for the point $(x, f(x))$.	$f(x) = a(x - h)^2 + k$ $5 = a(-1 - h)^2 + k$
Substitute values for the vertex (h, k) .	$f(x) = a(x - h)^2 + k$ $5 = a(-1 - (-3))^2 + 9$
Note: The order of the first two steps can be reversed.	
Solve for the variable a .	$5 = a(-1 - (-3))^2 + 9$ $5 = a(-1 + 3)^2 + 9$ $5 = a(2)^2 + 9$ $5 - 9 = a(4)$ $-4 = a(4)$ $a = -1$
Write the equation by substituting in the vertex and a value.	$f(x) = a(x - h)^2 + k$ $f(x) = -1(x - (-3))^2 + 9$

- c. Find a classmate who used a different point for $(x, f(x))$. Compare your equations. Explain why they are the same or why they are different.
Answers will vary. The equations are the same because both points are on the same parabola.



How can you be sure that $(-3, 9)$ is the vertex?



Why can any point on the parabola be used? Could the vertex be determined if the table of values only had the first three entries?



Pull a small group based on misconceptions from the Exploration. Use Guided Practice as an opportunity to scaffold connections between a table and necessary values for finding vertex.



Which points in the table helped you decide the vertex? What feature did these points have?

9.3.3 Guided Practice: Writing a Quadratic Function Using a Vertex and Point From a Table (continued)

2. A table of values for a quadratic function is shown.
- a. Identify the vertex of the quadratic function using the table.

Vertex $(\frac{2}{3}, -9)$

x	y
-1	16
$\frac{1}{3}$	-8
$\frac{2}{3}$	-9
1	-8
$\frac{7}{3}$	16

- b. Check your answer with your partner and discuss what strategy each of you used to identify the vertex. Correct your vertex if needed.

Answers will vary. I was able to see where the y values stop decreasing and began increasing again. I looked to see where the points began changing direction.

- c. Choose a point from the table, other than the vertex, to create the equation of the quadratic in the function.

$(1, -8)$

- d. Determine the a value of the quadratic using form $y = a(x - h)^2 + k$.

$$-8 = a\left(1 - \frac{2}{3}\right)^2 - 9$$

$$-8 = a\left(\frac{1}{3}\right)^2 - 9$$

$$-8 + 9 = a\left(\frac{1}{3}\right)^2$$

$$1 = a\left(\frac{1}{9}\right)$$

$$a = 9$$



Have students share their answers from question 2, part b, with the whole class. Allow groups to critique one another's methods and assess accuracy and reliability.



What if the point $(\frac{2}{3}, -9)$ was not listed in the table, how could you still determine the vertex?

9.3.3 Guided Practice: Writing a Quadratic Function Using a Vertex and Point From a Table (continued)

- e. Using the vertex and a value, write the equation of the function in vertex form.

$$y = 9\left(x - \frac{2}{3}\right)^2 - 9$$

- f. Rewrite the equation in standard form

$$y = ax^2 + bx + c.$$

$$y = 9x^2 - 12x - 5$$

3. A table of values for the function g is shown.

x	-11	-6	-1	4	9
$g(x)$	30	0	-10	0	30

- a. Determine the equation of the function g . Show your work.

$$y = a(x + 1)^2 - 10$$

$$0 = a(-6 + 1)^2 - 10$$

$$0 = a(-5)^2 - 10$$

$$0 + 10 = a(25)$$

$$10 = a(25)$$

$$a = \frac{2}{5}$$

- b. Give the equation in both vertex and standard form.

Vertex Form	$y = \frac{2}{5}(x + 1)^2 - 10$
Standard Form	$y = \frac{2}{5}x^2 + \frac{4}{5}x - \frac{48}{5}$



Encourage students to use fractions when expanding $\left(x - \frac{2}{3}\right)^2$. Students may want to change the fraction to a decimal or rely on a calculator. Build student understanding by using tape diagrams or other manipulatives.



Could the a -value be determined by just looking at the ordered pairs in the table?



If there are students struggling with converting from vertex to standard form, consider modifying the question for those students to get to vertex form and create additional opportunities for conversion practice.

9.3.4 Your Turn!

Component Overview: Students will work independently to write a quadratic equation in vertex and/or standard form given a table of values.



9.3.4 Your Turn!

- Write the quadratic equation in vertex form.

x	-10	-8	-2	4	6
$f(x)$	-29	-15	3	-15	-29

Vertex	Point
$(-2, 3)$	$(4, -15)$
Value of a	$y = a(x + 2)^2 + 3$ $-15 = a(4 + 2)^2 + 3$ $-15 = a(6)^2 + 3$ $-15 - 3 = 36a$ $-18 = 36a$ $a = -\frac{1}{2}$
Vertex Form	$y = -\frac{1}{2}(x + 2)^2 + 3$

- Write the quadratic equation in vertex and standard form for the quadratic function using the table.

x	-5	-3	-1	1	3
$f(x)$	55	7	-9	7	55

$$y = a(x + 1)^2 - 9$$

$$7 = a(1 + 1)^2 - 9$$

$$7 = a(2)^2 - 9$$

$$7 + 9 = 4a$$

$$16 = 4a$$

$$a = 4$$

Vertex Form	$y = 4(x + 1)^2 - 9$
Standard Form	$y = 4x^2 + 8x - 5$



For students doing well, challenge them to write question 1 in standard form as well. Be sure they show their work as part of the task.



Do you think the a -value will be positive or negative? How could you determine that?



What would you do if the table of values did not have a vertex displayed but showed the x -intercepts instead?

9.3.5 Wrap-Up: Ticket Out the Door

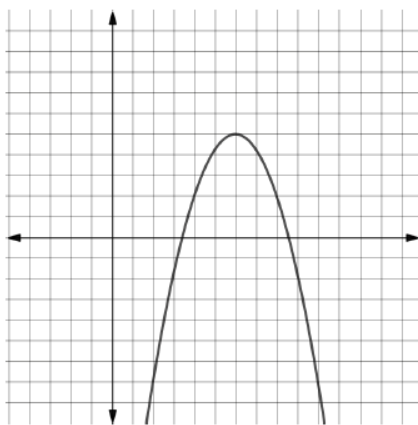
Component Overview: Students will complete a Ticket Out the Door to show their level of mastery of writing the equation of a quadratic function in vertex and standard form. This activity should be done independently, without assistance, so you can have data related to student performance on the concept of this lesson.



9.3.5 Wrap-Up: Ticket Out the Door

- Write the equation for the quadratic function in vertex and standard form.

x	$f(x)$
0	-22
2	-7
4	2
6	5
8	2



Look for students who use the point (4, 2) not (6, 5) as the vertex. These students have the misconception of using the ordered pair in the “middle” of the table instead of looking for a pattern to determine the vertex. You could ask students to circle which representation, the table, or the graph, they found to be most helpful in writing the equations.

$$2 = a(4 - 6)^2 + 5$$

$$2 = a(-2)^2 + 5$$

$$2 - 5 = a(4)$$

$$-3 = a(4)$$

$$a = -\frac{3}{4}$$

$$y = -\frac{3}{4}(x - 6)^2 + 5$$

$$y = -\frac{3}{4}(x^2 - 12x + 36) + 5$$

$$y = -\frac{3}{4}x^2 + 9x - 27 + 5$$

$$y = -\frac{3}{4}x^2 + 9x - 22$$

Vertex Form	$y = -\frac{3}{4}(x - 6)^2 + 5$
Standard Form	$y = -\frac{3}{4}x^2 + 9x - 22$